EPSR-test-ex109 Starting from scratch: no .pcof, no accumulators

Program(s): epsr | chained passes: 3 | verification: ALL IDENTICAL (all 3 passes)

What this case tests

Exercises the first `epsr` run of a new simulation [1,2,3]: only the `.EPSR.inp`, `.ato`, data and weights files exist, so default pair potentials, empty accumulators and every output file must be created. A second run tests atom types with zero coarse-graining value `natomsphere`.

Code exercised

Fortran (original, fortran_src/):

- `read_and_save_files.f`: `read_pcof` (missing file, `npairspcof` = `0` forces `ireset` = `iinit` = `1`, default pair potentials)
- `epsr_standalone_rev3.f`: `inita`, `initg` (no `.?01` files); `process_symbols.f`: `check_symbol` (caps `ncoeffp` at 1000)
- `ato_read_and_save.f`: `minitc` (7-value step-size line, `natomsphere` = `0`)

Rust (port, rust/epsr-core/src/):

- `pcof.rs`: `read_pcof`, `default_pair_potentials`; `symbols.rs`: `check_symbol`
- `structure.rs`: `inita`; `output.rs`: `initg`; `ato.rs`: `minitc`

Physical system

The EPSR-test system: a liquid mixture of 200 dimethyl sulphoxide (DMSO, (CH3)2SO) and 400 water molecules (DMSO mole fraction 1/3; 3200 atoms) at 300 K in a cubic box of side 33.12 Å (atomic number density 0.0881 atoms/Å^3). DMSO uses EPSR's Lennard-Jones + Coulomb reference potential with methyl rotational groups; water uses SPC/E Lennard-Jones and charge parameters [4] with EPSR's flexible intramolecular geometry (O-H 0.976 Å). The empirical potential is refined against three neutron diffraction data sets of deuterated DMSO in D2O, in null-scattering H2O/D2O and in H2O (ISIS SANDALS runs SLS03863, SLS03871, SLS03850), the isotope-substitution approach of Soper and Luzar [5]. Here the simulation starts with default reference potentials, `rminex` = `max(rminfac*dcor, roverlap)` and zero empirical potential.

Inputs

`scripts/make_inputs.sh` from the EPSR-test files: `.pcof`, `.EPSR.inpa`, `.erg`, `.g01`, `.y01`, `.f01`, `.s01` deleted; `.ato` step-size line with 7 values (7th = `tempdihreq`); `ncoeffp` `2000`, `ntimes` `2`, `niter` `2` in the `.EPSR.inp`. `scripts/clone_run.py` `nosph` and `scripts/zero_natomsphere.py` make the run `nosph` with `natomsphere` 0 for every type. `commands.txt`: both runs, `repeat` `3`.

How the accuracy was verified

`commands.txt` (2 program calls, repeated 3 times, each pass reading the files written by the previous one) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; only the date line of `.out` and the wall-clock columns of `fort.35` are masked). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in all 3 passes.
Pass 1 creates every output file (including `.pcof` and `.inpa`), passes 2 and 3 read them back. `ncoeffp` `2000` is capped to 1000 (the Fortran message prints it with i3 as `***`); the `.ato` is rewritten with a 10-value step line. The README reports gcov evidence for the `npairspcof.le.0` branch, the default pair potentials and `nwords.eq.7` in `minitc`.

Notes

In `nosph` `natomrcg` = `natomcg` = `0`, so the atom fractions stay un-normalised and several outputs contain NaN (e.g. `.erg` columns 6-8, `.r01`); this Fortran behaviour is reproduced, NaNs included. Default pair potentials do not set `efacp`.

References

[1] A. K. Soper, "Empirical potential Monte Carlo simulation of fluid structure", Chem. Phys. 202, 295-306 (1996).
[2] A. K. Soper, "Partial structure factors from disordered materials diffraction data: An approach using empirical potential structure refinement", Phys. Rev. B 72, 104204 (2005).
[3] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[4] H. J. C. Berendsen, J. R. Grigera and T. P. Straatsma, "The missing term in effective pair potentials", J. Phys. Chem. 91, 6269-6271 (1987).
[5] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).
[6] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.